

JONATHAN ONG CHONG AN

jonongca@gmail.com | +65 89205102 | jonongca.com
linkedin.com/in/jonongca | github.com/JonOng2002

Availability: January 2027 – August 2027

Singapore, SG

EDUCATION

Bachelor of Science, Information Systems (Product Development)

Jan 2023 – Dec 2027

Singapore Management University | Second Major: Computer Science (Artificial Intelligence)

SUMMARY

Information Systems undergraduate with hands-on experience building Python backend services, enterprise data pipelines, cloud infrastructure, and applied AI systems. Skilled in Python, FastAPI, SQL, PySpark, Docker, testing, and CI/CD, with a focus on reliability, debugging, and production-ready software.

EXPERIENCE

AI & Data Engineering Intern, Inland Revenue Authority of Singapore (IRAS)

May 2026 – Nov 2026

- Migrated 6 legacy SAS ETL scripts to PySpark for recurring batch workflows processing multi-million-row tax datasets, improving maintainability while supporting downstream analytics and reporting.
- Entrusted to lead the end-to-end integration of internal search engine APIs into an AI email assistant's RAG pipeline, owning backend retrieval logic and frontend integration through production rollout to improve access to trusted organisational knowledge.
- Contributing to the migration of an on-premises Python data ingestion pipeline to PySpark for Azure deployment, redesigning transformation logic and data flows to improve scalability, maintainability, and production readiness.
- Developed reusable validation checks across schemas, joins, timestamps, and business rules to detect migration discrepancies and strengthen confidence in PySpark pipeline outputs.
- Converted two form data processing workflows from batch Python scripts into reusable PySpark notebooks, enabling business users and data analysts to continue operational processing during migration while standardising CSV ingestion and ABFSS path configuration.
- Improved the reliability and observability of machine-learning APIs during migration to Azure ML by standardising model artefact management and strengthening logging, exception handling, and debugging workflows for developers.

DevOps Engineer Intern, YTL PowerSeraya

Jan 2026 – Apr 2026

- Built and rolled out Azure DevOps multi-stage YAML pipelines for 4 internal gas-management applications, reducing deployment time from 60 to 20 minutes and improving release consistency.
- Designed a self-hosted Azure DevOps agent architecture for secure IaaS deployments within an isolated VNet, enabling automated CI/CD execution on application VMs and creating reusable documentation for developer adoption.

Cloud Engineer Intern, Global Enterprise Mobility (GEM)

May 2025 – Aug 2025

- Built Jenkins CI/CD pipelines using Terraform-provisioned AWS infrastructure and GCP Secret Manager, reducing release time from 40 to 20 minutes while improving deployment repeatability and credential security.
- Implemented a DynamoDB idempotency layer to deduplicate webhook events, resolving a production defect affecting approximately 12–15% of transactions.

PROJECTS

PitchPerfect — AI Evaluation Platform | Product Owner & Data Pipeline Lead

Jun 2026 – Present

- Product Owner in a six-member Agile Scrum team developing an AI evaluation platform for the Lee Kuan Yew Global Business Plan Competition, reducing pitch-deck evaluation time by 80%.
- Lead the ingestion and retrieval workstream, translating a manual evaluation workflow into structured feature extraction, metadata processing, evidence retrieval, and readiness-assessment stages.
- Design the data pipeline to ingest labelled pitch decks, extract rubric-aligned features and metadata, generate embeddings, and retrieve supporting evidence for consistent and explainable evaluations.

YilongMa — Retrieval-Augmented Generation Data Pipeline

Aug 2025 – Dec 2025

- Built an end-to-end RAG pipeline using Python, ChromaDB, Llama 3.2 1B, and hybrid retrieval to transform transcript data into grounded responses from a searchable knowledge base.
- Processed 50+ hours of audio with Whisper, structured the transcripts for retrieval, and developed evaluation notebooks to assess retrieval relevance and response grounding.

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google IT Automation with Python | Java Certified Foundations Associate

SKILLS

Languages: Python, SQL, Java, JavaScript, Bash

Software Engineering: FastAPI, Flask, REST APIs, Pytest, Git, DSA

Cloud & DevOps: Docker, Azure, AWS, GCP, Terraform, Jenkins, CI/CD

Data & Databases: PostgreSQL, DynamoDB, PySpark, Pandas, Databricks, Azure Synapse, ADLS

AI/ML: RAG, Embeddings, Hybrid Retrieval, LLM Evaluation, Azure ML